Remote Sensing Changing Detection with Fake Changing Suppressing

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Abstract—**This document describes the most common article elements and how to use the IEEEtran class with LaTeX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.**

Index Terms—**Remote Sensing, Semantic Segmentation, Region Changing Detection, journal, LaTeX, paper, template, typesetting.**

I. Introduction

THIS is the introduction.

II. Related Work

Remote sensing change detection is closely related to visual representation learning, bi-temporal feature comparison, pseudo-change suppression, and dense prediction. This section reviews representative studies from these four aspects.

A. Visual Representation Learning for Remote Sensing

Remote sensing imagery contains complex spatial structures, spectral responses, and scale variations, which makes robust visual representation learning essential for downstream interpretation. Ji *et al.* proposed PASSNet [1] with a patch attention module for hyperspectral image classification. Han *et al.* studied optical remote sensing object detection using weakly supervised learning and high-level feature learning [2]. Wang *et al.* designed FSoD-Net [3] for full-scale detection in optical remote sensing imagery. Liu *et al.* investigated unsupervised contextual anomaly detection for remotely sensed data [4]. Xiao *et al.* introduced a diffusion-based super-resolution model for remote sensing images [5]. These studies improve remote sensing representation from classification, detection, anomaly modeling, and restoration perspectives, but they mainly operate on single images. In contrast, change detection requires discriminative representation as well as explicit comparison between two temporal observations.

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B. Bi-Temporal Remote Sensing Change Detection

Deep change detection methods usually learn bi-temporal representations with siamese or dual-stream architectures. Daudt *et al.* introduced fully convolutional siamese networks as an early deep comparison framework [6]. Fang *et al.* further improved siamese modeling through dense connections in SNUNet-CD [7]. Long *et al.* proposed SASiamNet [8] with self-adaptive feature interaction in a siamese framework. Peng *et al.* combined attention mechanisms with image differencing for change detection [9]. Wang *et al.* introduced attention-based deep supervision in ADS-Net [10]. Gu *et al.* focused on full-scale difference feature fusion in FDFF-Net [11]. Wang *et al.* proposed a high-resolution feature difference attention network for building change detection [12]. Chen *et al.* introduced transformer-based modeling for remote sensing change detection [13]. Ma *et al.* presented EATDer [14] with edge-assisted adaptive transformer detection. Li *et al.* further combined semantic flow and edge-aware refinement in SFEARNet [15] to improve efficient change reasoning. Although these methods have advanced bi-temporal fusion and boundary localization, many of them still treat visual discrepancy as the main evidence of change, which may lead to false responses in pseudo-change regions.

C. Pseudo-Change Suppression in Change Detection

Pseudo-change refers to visually apparent differences that are not annotated as semantic changes. It is commonly caused by illumination variation, seasonal transition, sensor inconsistency, atmospheric interference, or local imaging conditions. Zhang and Xue proposed SIAM-CDNET [16] to optimize edge detection and mitigate pseudo changes. Cheng *et al.* introduced a content-cleansing framework to remove unreliable cues before change inference [17]. Chen *et al.* formulated building change detection as an XOR-style binary relation problem [18]. Wang *et al.* proposed FIBTNet [19] to enhance feature interaction between bi-temporal observations. These methods share the view that pseudo-change cannot be fully handled by simple appearance differencing and requires stronger semantic or structural constraints.

Semi-supervised and contrastive learning methods also provide useful insights for pseudo-change suppression. Sun *et al.* explored semi-supervised change detection with siamese networks and graph attention [20]. Kondmann *et al.* proposed SemiSiROC [21] with an unsupervised teacher model. Wang *et al.* proposed reliable contrastive learning to separate changed and unchanged representations [22]. Wang *et al.* introduced STCRNet [23] with self-training and consistency regularization. These methods improve robustness by exploiting unlabeled data, consistency constraints, or representation separation. However, the core challenge of pseudo-change remains the inconsistency between visual difference and annotation semantics. Therefore, an annotation-consistent supervision strategy is still needed to guide the model toward benchmark-defined changes rather than all observable differences.

D. Dense Prediction and Semantic Segmentation in Remote Sensing

Dense prediction methods provide important design references for pixel-wise change detection, especially in foreground modeling and boundary delineation. As a semantic-segmentation reference, Xu *et al.* proposed RSSFormer [24] to enhance foreground saliency for remote sensing land-cover segmentation. Tang *et al.* developed WNet [25] with a hierarchical W-shaped structure for remote-sensing image change detection. Yuan *et al.* improved multi-scale feature modeling through the global filter in the frequency domain [26]. Lu *et al.* proposed a multi-scale feature progressive fusion network for change detection [27]. Ma *et al.* introduced edge-assisted adaptive transformer detection [14]. These methods show that accurate dense prediction benefits from foreground saliency, multi-scale context, and boundary-aware refinement. Nevertheless, semantic segmentation is usually single-temporal, whereas change detection must compare two temporal observations and distinguish semantic change from pseudo-change. Thus, segmentation-related techniques are useful components, but they do not by themselves solve annotation-consistent bi-temporal change detection.

III. Preliminary

A. Notations

To avoid ambiguity, the main symbols used in this paper are summarized in Table I. Their definitions are fixed throughout the remainder of the manuscript.

B. Region Changing Detection

Remote sensing change detection aims to identify whether each spatial location has undergone semantic change between two observations. Given a bi-temporal image pair $(\mathbf{I}^{t_1}, \mathbf{I}^{t_2})$, the task is to predict a binary change map $\hat{\mathbf{Y}} \in \{0,1\}^{H\times W}$, where 0 denotes unchanged and 1 denotes changed. In practical benchmark datasets, the target is not merely to respond to visual discrepancy, but to produce predictions consistent with the annotation criteria defined by the dataset.

C. Pseudo-Change

In remote sensing scenes, many regions exhibit visible appearance differences between two acquisition times even though the semantic land-cover category remains unchanged. These differences may be caused by illumination variation, seasonal transition, atmospheric interference, sensor difference, or local imaging conditions. When such regions are still annotated as unchanged in the ground truth, they form a pseudo-change phenomenon.

TABLE I
Summary of the main symbols used in this paper.

Symbol	Meaning
$\mathbf{I}^{t_1}, \mathbf{I}^{t_2}$	Bi-temporal input images captured at times t_1 and t_2.
$\hat{\mathbf{Y}}$	Predicted binary change map.
$\mathbf{Y}$	Ground-truth binary change map.
$\mathbf{Y}^e$	Ground-truth edge supervision map.
$\mathbf{P} \in \mathbb{R}^{2\times H\times W}$	Change prediction logits.
$\mathbf{P}^e \in \mathbb{R}^{2\times H\times W}$	Edge prediction logits.
$\mathbf{F}^{t_1}, \mathbf{F}^{t_2} \in \mathbb{R}^{C\times H\times W}$	Deep feature maps extracted from the two temporal observations.
$\mathcal{N}_\theta$	Change detection network parameterized by θ.
$\Omega = \{1,\dots,H\} \times \{1,\dots,W\}$	Spatial domain of an image.
$(i,j) \in \Omega$	Pixel location at row i and column j.
$p^1_{b,ij}, q^1_{b,ij}$	Foreground probabilities of the change and edge branches after softmax.
B	Batch size.
α	Balance factor between CE loss and IoU loss within the segmentation branch.
β	Weight of the contrastive branch in the final objective.
γ	Weight of the edge branch in the final objective.
m	Contrastive margin.
ε	Numerical stabilizer.
$\mathcal{L}_{\text{ce}}, \mathcal{L}_{\text{iou}}, \mathcal{L}_{\text{edge}}$ $\mathcal{L}_{\text{con}}, \mathcal{L}_{\text{seg}}, \mathcal{L}_{\text{total}}$	Cross-entropy, IoU, edge, contrastive, segmentation-fusion, and total optimization losses.

Formally, for a pixel location (i,j), let Δv_{ij} denote the visual discrepancy between the bi-temporal observations and let $y_{ij} \in \{0,1\}$ denote the benchmark annotation. A pseudo-change region satisfies

$$\Delta v_{ij} > 0, \qquad y_{ij} = 0, \tag{1}$$

which means that observable appearance variation exists, but the annotation semantics still indicate unchanged status.

Pseudo-change is one of the major causes of false positives in change detection. If a model is optimized only to amplify visual differences, it may over-respond to non-semantic variations and generate change predictions inconsistent with benchmark labels. Therefore, suppressing pseudo-change interference is essential for improving annotation-consistent change detection.

D. Segmentation Loss Computation

This work employs several standard supervised losses as basic optimization components. These losses are introduced here as preliminary definitions, whereas the proposed contribution lies in their bounded normalization and coupling with feature-level contrastive supervision in the Methodology section. Let $\Omega = \{1,\dots,H\} \times \{1,\dots,W\}$ denote the spatial domain of an image, and let $(i,j) \in \Omega$ denote the pixel at row i and column j. Here, $\mathbf{P}$ and $\mathbf{P}^e$ denote the logits of the change and edge branches, respectively, while $\mathbf{Y}$ and $\mathbf{Y}^e$ denote the corresponding binary supervision maps. For a mini-batch of size B, the foreground probabilities of the change and edge branches are given by

$$p_{b,ij}^{1} = \frac{\exp\left(\mathbf{P}_{b,1,i,j}\right)}{\sum_{c=0}^{1}\exp\left(\mathbf{P}_{b,c,i,j}\right)},$$
$$q_{b,ij}^{1} = \frac{\exp\left(\mathbf{P}_{b,1,i,j}^{e}\right)}{\sum_{c=0}^{1}\exp\left(\mathbf{P}_{b,c,i,j}^{e}\right)}. \tag{2}$$

For compactness, $\sum_{b,ij}$ denotes $\sum_{b=1}^{B}\sum_{(i,j)\in\Omega}$ in the following definitions. The cross-entropy loss for pixel-wise change classification is defined as

$$\mathcal{L}_{\text{ce}} = -\frac{1}{B|\Omega|}\sum_{b,ij}\log\frac{\exp\left(\mathbf{P}_{b,y_{ij}^{b},i,j}\right)}{\sum_{c=0}^{1}\exp\left(\mathbf{P}_{b,c,i,j}\right)}. \tag{3}$$

The foreground IoU loss for region-level overlap supervision is defined as

$$\mathcal{L}_{\text{iou}} = 1 - \frac{I_{\text{chg}}}{U_{\text{chg}} + \varepsilon}, \tag{4}$$

where

$$I_{\text{chg}} = \sum_{b,ij} p_{b,ij}^{1} y_{ij}^{b}, \qquad U_{\text{chg}} = \sum_{b,ij}\left(p_{b,ij}^{1} + y_{ij}^{b} - p_{b,ij}^{1} y_{ij}^{b}\right). \tag{5}$$

The foreground Dice loss for edge-aware supervision is defined as

$$\mathcal{L}_{\text{edge}} = 1 - \frac{2I_{\text{edge}} + \varepsilon}{S_{\text{edge}} + \varepsilon}, \tag{6}$$

where

$$I_{\text{edge}} = \sum_{b,ij} q_{b,ij}^{1} y_{ij}^{e,b}, \qquad S_{\text{edge}} = \sum_{b,ij}\left(q_{b,ij}^{1} + y_{ij}^{e,b}\right). \tag{7}$$

Here, $\varepsilon = 10^{-7}$ is a small constant for numerical stability. Since the implementation uses `num_classes=2`, the IoU and Dice losses are computed only on the foreground class.

IV. Methodology

A. Motivation

The existence of pseudo-change implies that reliable change detection requires both inter-temporal difference sensitivity and annotation-semantic consistency. In practical remote sensing datasets, visually changed regions may still be labeled as unchanged. Therefore, directly amplifying appearance discrepancy can lead to false positives and unstable generalization.

To address this issue, we design an annotation-consistency-oriented training strategy with two coupled components: (1) pseudo-change suppression in feature space via label-guided contrastive supervision, and (2) bounded multi-task joint optimization for segmentation, edge, and contrastive branches. This design shifts the objective from pure visual difference response to benchmark-consistent change prediction.

B. Overview

Fig. 1 illustrates the full training pipeline. Two temporal images are first encoded by a shared backbone, then fused by pyramid-level temporal interaction to produce change and edge prediction branches. In parallel, deep temporal features are extracted for the MarginalDistance process to form a label-guided contrastive supervision signal.

Given bi-temporal images $\mathbf{I}^{t_1}$ and $\mathbf{I}^{t_2}$, the network $\mathcal{N}_\theta$ outputs change logits $\mathbf{P}$, edge logits $\mathbf{P}^e$, and deep features $\mathbf{F}^{t_1}, \mathbf{F}^{t_2}$:

$$\left(\mathbf{P}, \mathbf{P}^{e}, \mathbf{F}^{t_1}, \mathbf{F}^{t_2}\right) = \mathcal{N}_\theta\left(\mathbf{I}^{t_1}, \mathbf{I}^{t_2}\right). \tag{8}$$

Following Fig. 1, supervision is organized into three branches: segmentation (CE and IoU), edge prediction, and contrastive discrepancy modeling. Their normalized terms are finally fused into a bounded objective for stable multi-task optimization.

C. Pseudo-Change Suppression via Label-Guided Contrastive Supervision

Let $(i,j)\in\Omega$ denote a pixel location, and let $\mathbf{F}^{t_1}, \mathbf{F}^{t_2} \in \mathbb{R}^{C\times H\times W}$ be deep features from the two temporal inputs. We first apply channel-wise L2 normalization:

$$\tilde{\mathbf{f}}_{ij}^{t_1} = \frac{\mathbf{F}_{ij}^{t_1}}{\left\|\mathbf{F}_{ij}^{t_1}\right\|_2}, \qquad \tilde{\mathbf{f}}_{ij}^{t_2} = \frac{\mathbf{F}_{ij}^{t_2}}{\left\|\mathbf{F}_{ij}^{t_2}\right\|_2}. \tag{9}$$

Then the pixel-wise Euclidean distance is

$$d_{ij} = \left\|\tilde{\mathbf{f}}_{ij}^{t_1} - \tilde{\mathbf{f}}_{ij}^{t_2}\right\|_2. \tag{10}$$

With binary label $y_{ij}\in\{0,1\}$ (0: unchanged, 1: changed), the pixel-wise contrastive penalty is

$$\ell_{\text{con}}^{ij} = \frac{1-y_{ij}}{2}d_{ij}^{2} + \frac{y_{ij}}{2}\max(0, m - d_{ij})^{2}, \tag{11}$$

where m is the contrastive margin. Unchanged pixels are pulled closer, while changed pixels are pushed apart up to the margin.

The image-level contrastive loss is

$$\bar{\mathcal{L}}_{\text{con}} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\ell_{\text{con}}^{ij}, \tag{12}$$

and its bounded form is

$$\mathcal{L}_{\text{con}}^{\text{norm}} = \text{clip}\left(\frac{\bar{\mathcal{L}}_{\text{con}}}{\max\left(1, \frac{1}{2}m^{2}\right)}, 0, 1\right). \tag{13}$$

D. Objective Function

Following Table I, let $\mathbf{Y}$ be the change label map, $\mathbf{Y}^e$ be the edge label map, and θ be network parameters. The segmentation branch combines CE and IoU losses:

$$\mathcal{L}_{\text{seg}} = \alpha\,\mathcal{L}_{\text{ce}}^{\text{norm}} + (1-\alpha)\,\mathcal{L}_{\text{iou}}^{\text{norm}}, \tag{14}$$

with

$$\mathcal{L}_{\text{ce}}^{\text{norm}} = 1 - \exp\left(-\mathcal{L}_{\text{ce}}\right), \tag{15}$$

$$\mathcal{L}_{\text{iou}}^{\text{norm}} = \text{clip}(\mathcal{L}_{\text{iou}}, 0, 1), \qquad \mathcal{L}_{\text{edge}}^{\text{norm}} = \text{clip}(\mathcal{L}_{\text{edge}}, 0, 1). \tag{16}$$

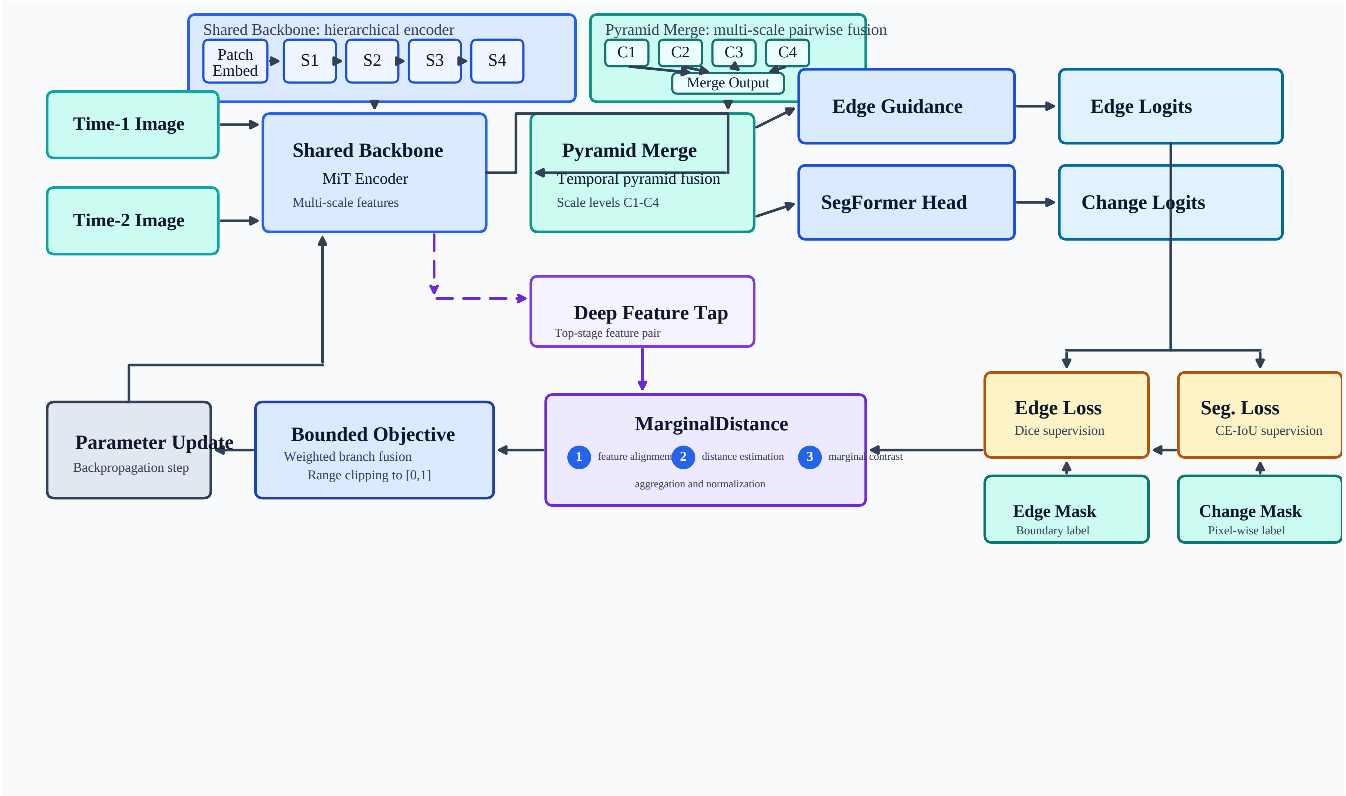

Fig. 1. Overall training pipeline of the proposed method. The model contains shared temporal encoding, pyramid-level fusion, dual prediction heads (change and edge), and a MARGINALDISTANCE branch for pseudo-change suppression. Multi-branch losses are bounded and fused to update network parameters.

The fused multi-task objective is

$$\mathcal{L}_{\text{total}} = (1-\beta-\gamma)\mathcal{L}_{\text{seg}} + \beta\mathcal{L}_{\text{con}}^{\text{norm}} + \gamma\mathcal{L}_{\text{edge}}^{\text{norm}}, \quad (17)$$

subject to $0 \leq \beta \leq 1$, $0 \leq \gamma \leq 1$, and $\beta+\gamma \leq 1$. The final bounded loss is

$$\mathcal{L} = \text{clip}(\mathcal{L}_{\text{total}}, 0, 1). \quad (18)$$

Accordingly, the optimization target is

$$\theta^* = \arg\min_{\theta} \mathbb{E}_{(\mathbf{I}^{t_1}, \mathbf{I}^{t_2}, \mathbf{Y}, \mathbf{Y}^e)} [\mathcal{L}] \,. \quad (19)$$

E. Algorithm: Annotation-Consistent Bounded Multi-Task Optimization

Algorithm 1 gives the complete optimization routine. The algorithm explicitly includes forward mapping, contrastive subroutine, bounded loss fusion, and gradient update. Variable semantics are explained inline at first appearance so the flow remains self-explanatory.

V. EXPERIMENT

Write the Experiment parts here.

A. Implementation Details

a) Datasets.: We evaluate on four widely-used change detection datasets: LEVIR-CD, WHU-CD, GZ-CD, and CLCD. These datasets cover various geographic regions, land cover types, and change patterns, providing a comprehensive testbed for cross-dataset generality.

b) Evaluation Metrics.: We report six metrics: Precision (Pre), Recall (Rec), F1-score (F1), Intersection over Union (IoU), Overall Accuracy (OA), and Kappa coefficient. These metrics capture different aspects of detection quality, including pixel-wise classification, region overlap, and agreement beyond chance.

c) Model Details.: The model details, training settings, and other implementation specifics will be described in the next sections.

d) Hyperparameters.:

e) Hardware Details.:

B. Main Results

C. Results on Pseudo-Change Datasets

D. Ablation Study

E. Visualization Analysis

F. Model Complexity and Efficiency

G. Discussion

VI. CONCLUSION

The conclusion goes here.

ACKNOWLEDGMENTS

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Algorithm 1 Annotation-Consistent Bounded Multi-Task Optimization

Require: bi-temporal images $\mathbf{I}^{t_1}, \mathbf{I}^{t_2}$; change labels $\mathbf{Y}$; edge labels $\mathbf{Y}^e$; contrastive margin m; CE/IoU balance α; branch weights β, γ; network parameters θ
Ensure: updated θ and final bounded loss $\mathcal{L}$

Perform forward mapping to obtain $\mathbf{P}$, $\mathbf{P}^e$, $\mathbf{F}^{t_1}$, and $\mathbf{F}^{t_2}$ by Eq. (8)
Compute $\mathcal{L}_{\text{con}}^{\text{norm}}$ using MARGINALDISTANCE($\mathbf{F}^{t_1}, \mathbf{F}^{t_2}, \mathbf{Y}, m$)
Compute $\mathcal{L}_{\text{ce}}$, $\mathcal{L}_{\text{iou}}$, and $\mathcal{L}_{\text{edge}}$ as defined in Section III
Bound branch losses: $\mathcal{L}_{\text{ce}}^{\text{norm}}$ by Eq. (15); $\mathcal{L}_{\text{iou}}^{\text{norm}}$ and $\mathcal{L}_{\text{edge}}^{\text{norm}}$ by Eq. (16)
Compute $\mathcal{L}_{\text{seg}}$ by Eq. (14)
Compute $\mathcal{L}_{\text{total}}$ by Eq. (17)
Compute final bounded loss $\mathcal{L}$ by Eq. (18)
Update θ via backpropagation with objective $\mathcal{L}$

procedure MARGINALDISTANCE($\mathbf{F}^{t_1}, \mathbf{F}^{t_2}, \mathbf{Y}, m$)
  Define pixel-coordinate set Ω
  for each $(i, j) \in \Omega$ **do**
    Normalize $\mathbf{F}_{ij}^{t_1}$ and $\mathbf{F}_{ij}^{t_2}$ to unit vectors by Eq. (9)
    Compute feature distance d_{ij} by Eq. (10)
    Compute pixel penalty ℓ_{con}^{ij} by Eq. (11) with label $y_{ij} \in \{0, 1\}$
  end for
  Average over pixels to obtain $\bar{\mathcal{L}}_{\text{con}}$ by Eq. (12)
  Normalize and clip $\bar{\mathcal{L}}_{\text{con}}$ to $\mathcal{L}_{\text{con}}^{\text{norm}}$ by Eq. (13)
  return $\mathcal{L}_{\text{con}}^{\text{norm}}$
end procedure

APPENDIX

PROOF OF THE ZONKLAR EQUATIONS

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